

A Sodium Germanosilicide with Unusual Network Topology

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ABSTRACT: The germanosilicide $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ ($0.4 \leq x \leq 1.1$, $4.7 \leq y \leq 9.3$) was synthesized under high-pressure, high-temperature conditions. The novel guest–host compound comprises a unique tetrel framework with dual channels housing sodium and smaller, empty $(\text{Si},\text{Ge})_9$ units. The arrangement represents a new structure type with an overall structural topology that is closely related to a hypothetical carbon allotrope. Topological analysis of the structure revealed that the guest environment space cannot be tiled with singular polyhedra as in cage compounds (e.g., clathrates). The analysis of natural tilings provides a convenient method to unambiguously compare related tetrel-rich structures and can help elucidate new possible structural arrangements of intermetallic compounds.

Intermetallic compounds are typically ordered solids composed of two or more metals or semimetals, and the properties observed are often superior to the individual constituents.^{1,2} For the case of silicon and germanium, a large number of binary intermetallic phases are known to exhibit large structural variety including simple close-packed arrangements, polyanionic Zintl phases, and more complex extended clathrate and open-framework networks hosting guest atoms.^{3,4} This structural variety yields diverse physical properties that are important for a broad range of applications including catalysis,⁵ solar-thermal applications,⁶ thermoelectrics,^{7,8} and superconductors.⁹ While overarching principles that govern structural arrangements, stabilities, and physical properties are still being developed, the discovery of novel intermetallic compounds can facilitate a broader understanding of complex bonding interactions and open new possibilities for applications.^{10,11}

High-pressure, high-temperature methods have proven to be versatile tools to access new compositions and structure types with exceptional properties.¹² Under compression, a volume reduction and, therefore, a densification of matter occurs, resulting in increased electrostatic interactions.¹³ For the case of group-14 elements, a wide range of novel guest–host framework types and compositions have been synthesized under high-pressure conditions, in addition to the clathrate structures that are formed at low pressure.^{14–29} Many framework types exhibit variable guest atom occupancies, and thus tunable electronic structures, and in some cases, guest atoms or intercalants may be topotactically removed to produce novel tetrel allotropes.^{30–34} The exploration of higher-order systems with the possibility of multipartite frameworks adds additional degrees of freedom to expand the structural diversity of 3D frameworks.

Herein, we report the synthesis of the novel compound $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$. The new structure comprises a 3D network of four- (4b)Si/Ge and three-bonded (3b)Ge atoms that

incorporate Na into extended cavities (Figure 1). The Si:Ge ratio is tunable based on the starting composition. The cavities share similarities with other tetrel-rich compounds and a topological analysis of natural tilings allowed for the structural classification among the family of tetrelides.

The new compound, $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ ($0.4(1) \leq x \leq 1.1(1)$, $4.7(2) \leq y \leq 9.3(8)$), was obtained by high-pressure, high-temperature synthesis (7.5 GPa, $500 \leq T \leq 650$ °C for 1 to 3 h) from mixtures of the elements (synthesis procedure detailed in the Supporting Information). Recovered samples consist of gray pellets comprised of $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ intergrown with Si/Ge alloy and an air-sensitive, sodium-rich phase (based on energy-dispersive X-ray spectroscopy, EDS, see Table S9). Crystals of the target phase can be isolated from crushed product agglomerates and persist after washing.

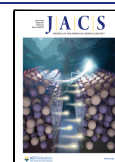
X-ray diffraction measurements reveal that $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ crystallizes in space group $C2/m$ with $a \approx 10.70$ Å, $b \approx 3.96$ Å, $c \approx 10.73$ Å and $\beta \approx 117.7^\circ$ (Table S1). The final structural model was determined using single-crystal X-ray diffraction data (SXRD) obtained with synchrotron radiation (Advanced Photon Source, sector 13 ID-D). The refinement resulted in five atomic positions Na1, Si1, Si2, Si3, and Ge4. All Si positions exhibit mixed Si/Ge occupancy with the largest Ge fraction on Si1, in line with longer distances compared to contacts involving the Si2 and Si3 positions (Tables S6 and S8). No mixed Si/Ge occupancy was observed for Ge4, and the (3b)Ge–(3b)Ge distances (Tables S6 and S8) fall into the range of (3b)Ge–(3b)Ge distances in other alkali, alkaline

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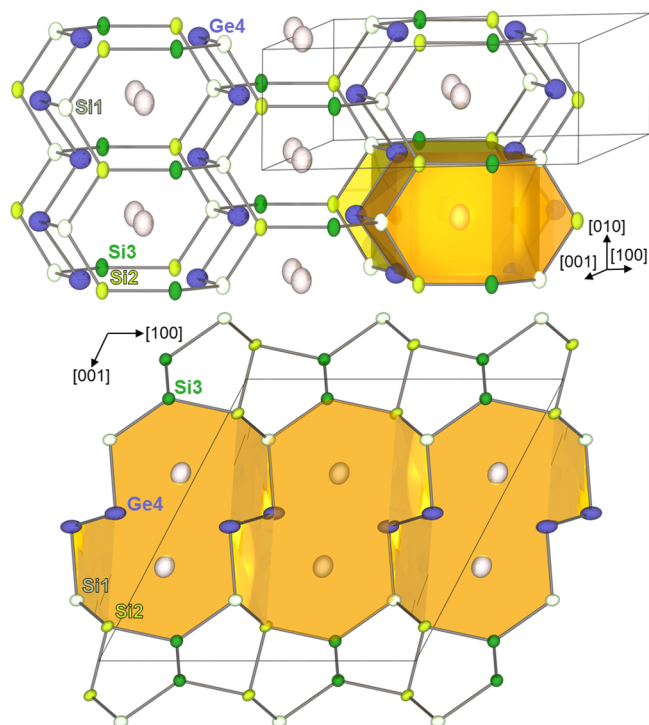


Figure 1. Crystal structure of $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ with a network of (3b) Ge and (4b) Si/Ge forming dual channels (orange) housing two Na atoms and empty (Si,Ge)₉ units.

earth, and rare earth germanides.³⁵ NaI showed partial occupancy (the refinement procedure is detailed in the SI). The unit cell volumes refined from powder X-ray diffraction data (PXRD) increase with increasing Ge content in the precursor mixture (Figure S1, Table S1), indicating a larger amount of Ge substitution within the structure. The refined Ge:Si ratio for all samples varies between ~ 0.5 – 1.5 in agreement with EDS and wavelength-dispersive X-ray spectroscopy (WDS) (Figures S1, S2 and Tables S1–S3).

The structural arrangement consists of a 3D network of (3b) Ge and (4b) Si/Ge that form cassinoidal-shaped cavities housing two Na atoms (Figure 1). These cavities form extended channels along [010] that are alternatingly offset along [100]. Along [001], one row of filled cavities is followed by puckered layers of hexagonal Si/Ge-rings forming empty (Si,Ge)₉ units. Interestingly, similar puckered layers and empty units are found in ternary P/Si and As/Si frameworks despite nonfour-bonded pentel atoms.^{36–38} The host lattice arrangement is closely related to a hypothetical 3D carbon allotrope that belongs to a class of topological nodal-line semimetals,³⁹ called *m*-C₈, reported in the carbon allotrope database SACADA as 3,4³T71.⁴⁰ The only difference between the frameworks is that three-center nodes (3b) of $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ are pyramidal rather than flat as in the C allotrope.

The stacking arrangement of filled cavities in straight columns along [010] resembles stacking sequences in MgSi_5 ,¹⁸ LaSi_{10} ,¹⁴ $\text{Na}_4\text{Ge}_{13}$,⁴¹ and MT_6 ($M = \text{Na}, \text{Sr}, \text{Ba}, \text{Eu}, \text{Tl} = \text{Si}, \text{Ge}$).^{16,19,42–44} Whereas (4b) Si/Ge⁰ is the usual case for tetrel-based frameworks,⁴⁵ examples with (3b) Ge are known (see Table S11). Interestingly, three-bonded vertices are found in BaGe_{6-x} ,¹⁶ SrGe_6 ,¹⁷ $\alpha\text{-LaSi}_5$,¹⁴ LaGe_5 ,¹⁵ and BaGe_5 ²⁰ on opposing sides of filled cavities, as is observed in $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$. Seemingly, structures containing three-bonded species are associated with the occurrence of larger guest

environments as compared with those comprised exclusively of four-bonded species.

To unambiguously classify the new germanosilicide among other 3D silicon- or germanium-based, guest–host frameworks, we performed a topological analysis of natural tiles⁴⁶ using ToposPro.⁴⁷ Whereas the crystal–chemical description of structural motifs relies on subjective interpretations,⁴⁸ the topological analysis utilizes a discrete set of rules to describe the tiling of 3D space.^{46,49} Tetrel-rich (MT_x , $x \geq 2$) alkali, alkaline earth, or rare earth metal compounds were compiled from the Inorganic Crystal Structure Database (ICSD),⁵⁰ resulting in 117 compounds with 20 different structure types in total, and their structures were decomposed into natural tilings for topological analysis (Figure 2, Table S11).

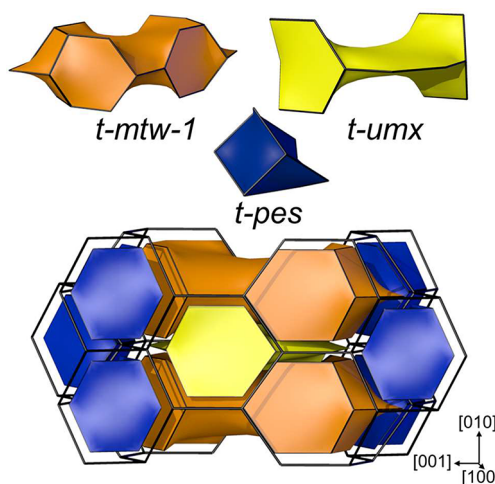


Figure 2. Framework tiling of $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$. The guest environment is represented by a combination of *t-mtw-1* and *t-umx* tiles.

The topological analysis revealed that the guest environments for these silicon- or germanium-rich frameworks can be either described by singular polyhedral tiles, or subdivided into multiple tile elements to fill 3D space (see Table S11). Single-tile polyhedral guest environments are defined as cages, as in clathrate compounds such as $\text{Na}_8\text{Si}_{46}$.⁵¹ Other compounds, such as MgSi_5 , LaSi_{10} , and type-VIII clathrates also contain single-tile guest environments, and therefore may be considered as cage-based compounds, but they also include additional nonguest-environment tiles. Another category of compounds investigated contains guest environments that are not unambiguously represented by one type of tiling. These types of guest environments are observed in several structures such as MT_6 ($M = \text{Na}, \text{Sr}, \text{Ba}, \text{Eu}, \text{Tl} = \text{Si}, \text{Ge}$), and cannot be considered as cages due to their multitile nature. These types of structures are thus generally defined as open-framework compounds that contain cavities. Finally, several analyzed structures contain mixed motifs with both cage (single-tile) and cavity (multitile) guest environments. Cage-based structures exhibit the largest distances between guest atoms and the polyhedra generally have smaller maximum face ring sizes as compared with structures possessing noncage (cavity) guest environments. The maximum ring sizes associated with the tiling elements for guest environments can serve as an additional measure for the classification of 3D networks as cage versus open-framework compounds (Figure 3).

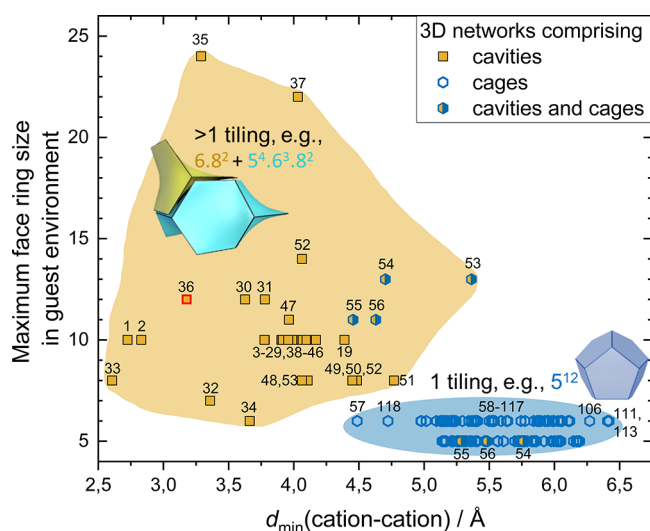


Figure 3. Minimum guest–guest distances vs maximum face ring size (number of ring elements) for unique guest environments in tetrel-rich alkali, alkaline earth, and rare earth metal silicides/germanides (see Table S11). $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ (No. 36) is outlined in red.

The guest environments within $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ cannot be represented by singular polyhedral tiles and consist instead of two tile types, *t-mtw-1* and *t-umx*, both found in zeolites (Figure 2, Table S11).⁴⁸ Therefore, $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ is classified as an open-framework compound, which possesses cavities with shorter guest–guest distances and larger ring sizes compared to cage-based structures. The new compound also contains empty (Si,Ge)₉ units represented by *t-pes* tiles, as found in the zeolite EEI and the high-pressure compounds NaSi_6 ,¹⁹ SrGe_{6-x} ,⁵² and SrGe_6 ,¹⁷ as well as *allo-Si*,³⁰ *allo-Ge*,^{31,53–55} and $\text{Li}(\text{Na})_7\text{Ge}_{12}$.⁵⁶ The novel framework is represented by a $3,4^3\text{T}71$ net according to the TopCryst system.⁵⁷ The underlying net can be related to hypothetical zeolites with *sqc2060*⁵⁸ (corresponding to Deem's⁵⁹ PCOD 8126237) and *4,4,4T3425-HZ*⁵⁷ topology (Deem's PCOD 8045767), neither of which have been observed experimentally before. As the transformation to the first net comprises an intermediate $3\text{D } 3^2,4^2$ -coordinated net without change in space group, a solid-to-solid transformation becomes possible. Transformation to the latter topology requires an intermediate *honeycomb* net and, therefore, a decrease in net periodicity, pointing toward amorphization rather than structural transformation. Strikingly, no straightforward transformation pathway between the $3,4^3\text{T}71$ net and common 4-coordinated tetrel nets exists, highlighting the unusualness of the new tetrel network (see SI for details). However, topologically similar networks are derived from slicing and rearranging diamond-type Si, highlighting the potential for further possible structural richness.⁶⁰

Given the open-framework nature of $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$, Na mobility and guest atom removal are conceivable, similar to the formation of Si_{24} from NaSi_6 ³³ and predicted desodiation of Na_4Ge_4 , $\text{Na}_{1-x}\text{Ge}_{3+z}$ and $\text{Na}_x\text{Ge}_{136}$.⁶¹ The theoretical storage capacity for the idealized stoichiometry $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ was calculated based on Faraday's law to be 149 mAhg^{-1} , which is similar to sodium battery materials such as NaTiO_2 .⁶² Despite the reduction of the Na content that was observed after samples were kept at ambient conditions for several weeks (Figures S8–S10 and Tables S9 and S10) and upon heating to 100°C (Figures S4–S7), the observed Na removal is

accompanied by a broadening of PXRD reflections (Figure S7, S10) and an increase in the (Si,Ge) alloy content of the sample, indicating a degradation of the crystallites and the 3D structure, as well as disproportionation (see SI).

Density functional theory (DFT) calculations were performed on an idealized model $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ without Ge substitution on (4b)-framework sites. At 10 GPa, this model structure is located on the ternary convex hull when considering known binary and ternary compounds in the Na–Si–Ge system (Figure 4), indicating that the new

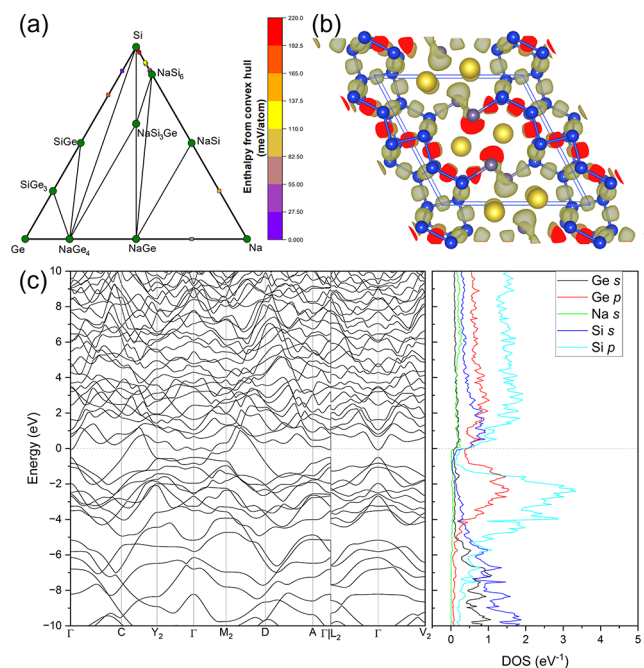


Figure 4. (a) Calculated Na–Si–Ge phase diagram at 10 GPa. Green circles indicate stable structures on the convex hull. Squares represent metastable structures colored by their hull distance. (b) ELF isosurface for $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ where olive coloring represents $\text{ELF} = 0.8$. (c) Electronic band structure of $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ with density of states (DOS) contributions projected onto atomic orbitals.

compound is likely thermodynamically stable under high-pressure conditions. At ambient pressure, $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ remains dynamically stable with a formation enthalpy of ca. -153 meV/atom , but is metastable with respect to known binaries (such as Na_4Ge_4 ,⁶³ and $\text{Na}_8\text{Ge}_{46}/\text{Na}_x\text{Ge}_{136}$).⁶⁴

According to the Zintl–Klemm concept, the occurrence of (3b)Ge and (4b)Si/Ge results in an electron balance of $\text{Na}^{+}_{4-x}[(3b)\text{Ge}^{-}]_y[(4b)\text{Si}^0]_{16-y}$. Nevertheless, the calculated band structure for the idealized composition $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ shows metallic behavior with a finite density of states (DOS) at the Fermi level, pointing to a deviation from the idealized charge-balanced picture. The DOS plot shows significant contributions from the Ge *p* and Si *p* orbitals at the Fermi level, with Ge showing a more dominant role (Figure 4).

To understand this apparent contradiction, we probed the electronic structure through Electron Localization Function (ELF) and Bader charge analysis. Bader analysis revealed noticeable charge transfer from Na atoms to the Ge–Si framework. Notably, Ge atoms accumulate more charge than Si. This is consistent with the ELF results; the extra electron of the 3b Ge forms a lone pair, exhibiting sp^3 bonding characteristics. Additional crystal orbital Hamilton population

(COHP) analysis provides detailed and atom-specific insights into the bonding nature of states within a particular energy window. For $\text{Na}_4\text{Ge}_4\text{Si}_{12}$, it is evident that a significant number of antibonding states are occupied below the Fermi level (Figure S13), which is directly correlated with the occupied conduction bands and the metallic nature of the compound. This picture shows similarities to SrGe_6 , which exhibits more complex guest–host bonding interactions compared with other open-framework compounds comprising cavities.¹⁷ Given that the idealized $\text{Na}_4\text{Ge}_4\text{Si}_{12}$ stoichiometry was never observed experimentally, the complete picture involving partial Na occupancy and lattice substitutions is more complex.

In summary, a new type of Si/Ge framework was obtained by high-pressure, high-temperature techniques. The new guest–host compound $\text{Na}_{4-x}\text{Ge}_y\text{Si}_{16-y}$ ($0.4 \leq x \leq 1.1$, $4.7 \leq y \leq 9.3$) contains ellipsoidal cavities that house Na, while smaller empty (Si,Ge)₉ units complete the 3D space-filling arrangement. Topological analysis of natural tilings revealed a structural relationship with a hypothetical carbon allotrope and allowed for the classification among tetrel-rich silicides and germanides. The analysis revealed that guest environments comprising cages can be described by a single type of tiling, whereas open-framework structures require more than one type to describe individual guest environments. Following the approach to predict new zeolite frameworks, the analysis of structural topology can be employed in the search for new intermetallic frameworks.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c03960>.

Details about synthesis, methods, as well as analyses results including powder and single crystal X-ray diffraction, X-ray spectroscopy, thermal analysis, crystal chemical and topological analysis, and theoretical calculation details (PDF)

Accession Codes

CCDC 2354921 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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